II-Probabilistic Approaches Inference

Foundations of Neuro-Symbolic AI

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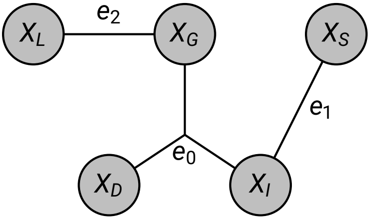
Outline

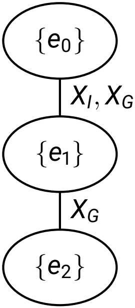
- Variable Elimination
- Approximative inference

Contraction by sub-contractions

Idea: Close one variable after another in a contraction.

Example: Student Markov network





Tree width

Size of the largest cluster is the "bottleneck".

Hardness of contractions

Finding the best order of the variables to be contracted is NP-hard, since related to the triangulation problem of a graph.

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- Approximative inference

Approximation schemes

Particle based:

- Draw samples from a distribution (or an approximation of it).
- Estimate contractions based on contraction with the corresponding empirical distribution.

Message passing based:

- Ignore loops in the graph and do message-passing schemes circulating in the network.
- Use converged messages as estimates of local means.

Forward sampling

Given a Bayesian network, sample iteratively the children given the parents.

Markov chain monte carlo

To sample from some hard distributions:

- Design a Markov chain converging to the desired distribution

Metropolis Hastings algorithms

Example: Gibbs sampling.